

PROJECT 2 · AUDIO FOUNDATION MODELS

Audio Foundation Models for Acoustic Scene Classification

A minimal demo: load audio, extract a **WavLM** embedding, train a tiny classifier,
predict the scene.

Load audio → WavLM → classifier → prediction

THE TASK

What are we classifying?

Given a short audio clip, predict the **acoustic scene** — the environment where it was recorded.



Restaurant

cafe chatter, cutlery



Street

traffic, crowds



Office

keyboards, quiet talk



Car

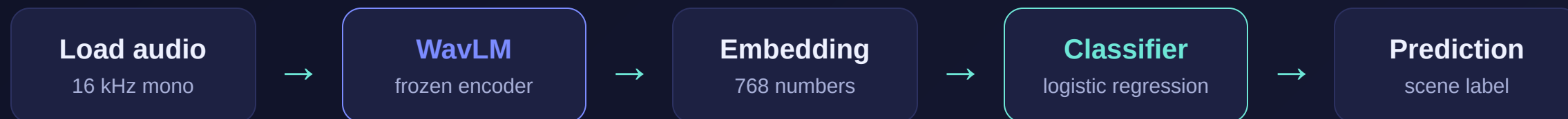
engine, road noise

This is a classic **acoustic scene classification** problem — useful for context-aware apps, smart assistants, and robotics.

THE APPROACH

Foundation model + linear probe

We do **not** train a deep network from scratch. WavLM is pre-trained on huge audio data, frozen, and used as a feature extractor.



Key idea: the expensive model only runs forward passes; the part we train is tiny and fast.

THE MODEL

WavLM from Hugging Face

WHAT	DETAILS
Model	<code>microsoft/wavlm-base-plus</code>
Type	Self-supervised speech/audio foundation model (Microsoft)
Input	Raw waveform at 16 kHz
Output	Sequence of 768-dim frame vectors → mean-pooled to one vector per clip
Library	<code>transformers</code> — one-line load with <code>WavLMModel</code>

HuBERT swap: change repo to `facebook/hubert-base-ls960` and use `HubertModel` — same pipeline.

TUT Acoustic Scenes 2017 — from Hugging Face

Dataset

`Mahia/TUT2017` mirrors the TUT 2017 acoustic scenes benchmark.

Our labels map to dataset classes:

Restaurant ← `cafe/restaurant`

Street ← `city_center`

Office ← `office`

Car ← `car`

Lightweight download

Full dataset is ~12 GB. For the demo we:

- Download the label CSV (~few KB)
- Pick ~20 clips per scene
- Fetch **only those files** via `hf_hub_download`

Eight clear steps

1

Install dependencies

2

Configure scenes &
model

3

Download clips from HF

4

Load & resample audio

5

Load WavLM + embed
fn

6

Extract all embeddings

7

Train classifier +
evaluate

8

Predict one clip

Notebook: `notebooks/audio-foundation-models/acoustic_scene_classification.ipynb`

HOW EMBEDDING WORKS

From waveform to vector

```
inputs = feature_extractor(wav, sampling_rate=16000, return_tensors="pt")
frames = wavlm(**inputs).last_hidden_state    # (1, time, 768)
embedding = frames.mean(dim=1)                # mean-pool → (768,)
```

- `AutoFeatureExtractor` converts the raw wave into model inputs.
- WavLM outputs one hidden vector per time frame.
- **Mean-pooling** over time gives a single clip-level embedding.
- Same embedding feeds both training and inference.

Linear probe with logistic regression

Training

- Standardize embeddings (`StandardScaler`)
- Fit `LogisticRegression` on the embeddings
- Split **by recording**, not randomly — avoids leakage

Evaluation

- Group-aware cross-validation
- Every clip tested on an **unseen recording**
- Accuracy, report & confusion matrix

With only ~80 clips, a frozen WavLM + linear head reaches **~91%** on held-out recordings — an honest score with no leakage.

Predicting a new clip

```
pred, proba = predict_scene("some_clip.wav")  
  
true scene : Office  
predicted  : Office  
confidence : {'Office': '87.2%', 'Restaurant': '8.1%', ...}
```

The `predict_scene()` helper runs the full pipeline on any `.wav` file and returns the label plus per-class probabilities.

TAKEAWAYS

What this demo shows

- **Audio foundation models** (WavLM / HuBERT) make strong feature extractors out of the box.
- A **linear probe** on frozen embeddings is enough for a working scene classifier.
- Both **model and dataset** come from the Hugging Face Hub — no manual data prep.
- The notebook is intentionally **simple and explained** — a clear teaching/demo artifact.

EXTENSIONS

How to go further

More data

Increase `N_PER_CLASS` or download the full TUT dataset for higher accuracy.

Try HuBERT

Swap the model repo and compare embeddings side by side.

Better pooling

Use attention pooling or multi-window averaging instead of a single mean.

Fine-tune

Unfreeze WavLM and train end-to-end for the best possible accuracy.

THANK YOU

Questions?

Notebook: `acoustic_scene_classification.ipynb`

Slides: `acoustic-scene-classification.pdf`

Model: `microsoft/wavlm-base-plus` · Data: `MahIA/TUT2017`

[Run the notebook top to bottom](#)